Course Project

A core objective of this course is to apply machine learning principles to solve real-world problems. In this project, you will apply the machine learning principles learned in this course to solve a real-world problem. You will build different models to analyze complex data and make accurate predictions.

Project Requirements

You will work on two machine learning projects from the Cybersecurity domain:

1. Network Intrusion Detection (Phase 1)

- **Dataset:** A dataset simulating a wide range of intrusions in a military network environment. The dataset consists of 31 quantitative and qualitative features including the target column "class".
- **Task:** Develop four distinct classification ML models and implement one ensampling technique to classify network connections as either 'normal' or 'anomaly' based on these attributes. Anomalous connections indicate potential intrusions or malicious activities

- **Challenge:** Address the class imbalance issue by employing appropriate techniques such as under-sampling or over-sampling.

2. Kaggle Competition (Phase 2)

- **Dataset:** the dataset is in the same domain of cybersecurity.
- **Task:** Prepare for a Kaggle competition by building and refining different regression models on a **soon-to-be-provided cybersecurity dataset**.
    - Develop at least four distinct regression models.
    - Evaluate and compare their performance.
    - Submit your best regression model's to Kaggle.

By completing these projects, you will gain hands-on experience in applying machine learning techniques to real-world financial problems, including data preprocessing, model selection, training, and evaluation.

Project Phases

Phase I [10 Marks]: Data Exploration and Model Implementation

Task 1: Data Exploration and Preprocessing

1. **Data Loading and Cleaning:** Load the datasets and handle missing values, outliers, and inconsistencies.
2. **Exploratory Data Analysis (EDA):**
   - Visualize data distributions using histograms, box plots, and scatter plots.
   - Identify correlations between features.
   - Explore potential relationships between features and the target variable.
3. **Feature Engineering:**
   - Create new features or transform existing ones if necessary.
   - Handle categorical variables using appropriate encoding techniques.
   - Choose 20 most important features to work with.
   - Scale numerical features if required.

Task 2: Model Implementation

1. **Implement 4 classification models,** ensuring each model is applied to the most appropriate dataset type. you can use these models but you are not limited to any of them.
   - K-Nearest Neighbors (KNN)
   - Logistic Regression
   - SVM
   - Random Forest
   - Decision Tree

2. **Implement 1 ensembling technique,** combining the predictions of multiple models to make a stronger, more reliable prediction (e.g. Voting).

3. **Model Training and Evaluation:**
   - Train each model on the prepared dataset.
   - Evaluate model performance using appropriate metrics (e.g., accuracy, precision, recall, F1-score, MSE, RMSE, MAE).
   - Compare the performance of different models and select the best-performing one.

Task 3: Report

The report must include the following:

- **Problem Statement:** Clearly define the problem and its significance.
- **Data Exploration:** Summarize key insights from the exploratory data analysis.
- **Data Preprocessing:** Describe the preprocessing steps taken, including handling missing values, outliers, and feature engineering.
- **Model Selection and Implementation:** Explain the rationale for choosing specific models and provide details on the implementation.
- **Model Evaluation:** Present the performance evaluation results for each model, including metrics and visualizations.
- **Conclusion:** Summarize the findings and discuss the limitations of the models.

Phase I Performance Evaluation and Grading:

Your performance in this phase will be evaluated based on the following:

1. **Code Quality:** Clarity, efficiency, and documentation.
2. **Model Performance:** Accuracy and generalizability of the models.
3. **Data Analysis:** Depth of analysis and insights derived from the data.
4. **Report Writing:** Clarity, organization, and presentation of results.
5. **Code Explanation:** Ability to explain the code and the rationale behind design choices.

Phase I List of Deliverables:

1. .Pdf file for the Report
2. .py file for the code
3. A clear task division for each member of the team

Phase I deadline: 15 April 2025, 11:59 PM

Phase II[10 Marks]: Kaggle Competition and Model Refinement

Task:

Participate in a Kaggle-style competition using the provided dataset. The goal is to develop a machine learning model that outperforms your colleagues' submissions.

By participating in this Kaggle-style competition, you will gain valuable experience in applying machine learning techniques to real-world problems, collaborating with teammates, and Iteratively refining models to achieve optimal performance.

Requirements:

- **Model Implementation:**
  - Implement at least four different ML models, exploring diverse algorithms (e.g., linear regression, decision trees, SVMs, Random Forests, neural networks).
  - Utilize advanced techniques (e.g., hyperparameter tuning, ensemble methods, feature engineering) to optimize model performance.
  - Consider alternative model architectures or algorithms that may be better suited to the problem and dataset.
- **Performance Evaluation:**
  - Utilize appropriate evaluation metrics (e.g., accuracy, precision, recall, F1-score, RMSE, MAE) to assess models' performance.
  - Conduct a thorough analysis of each model's strengths and weaknesses, explaining their performance differences.
- **Model Submission:**
  - Submit a single and optimized model to the Kaggle competition.
  - Ensure the model is well-documented, clearly explaining the chosen algorithm, preprocessing steps, and any feature engineering or model tuning techniques..
- **Report:**
  - **Problem Statement**
    - Clearly define the problem you are solving, its significance in the cybersecurity domain, and the potential impact of an effective solution.

- **Data Exploration and Preprocessing**
  - Summarize key insights from exploratory data analysis (EDA), including visualizations and statistical summaries.
  - Describe the preprocessing steps taken, including handling missing values, outliers, feature scaling, encoding categorical variables, feature engineering, and feature selection.
- **Model Selection and Implementation**
  - Justify the choice of the final model submitted to Kaggle, comparing its performance and characteristics to other models considered.
  - Explain the specific techniques used for model improvement, including hyperparameter tuning, ensemble methods, and any innovative approaches.
- **Model Evaluation**
  - Present the performance evaluation results for all models, including the chosen metrics, confusion matrices, and visualizations.
  - Discuss the final model's limitations and potential areas for further improvement.
- **Conclusion**
  - Summarize the key findings and lessons learned from the project.
  - Discuss the potential real-world impact of the model, its limitations in the context of cybersecurity, and ethical considerations.

Phase II Performance Evaluation and Grading:

Your performance in this phase will be evaluated based on the following:

- **Model Performance:** Your model’s ranking on the Kaggle leaderboard will be a **significant factor.**
- **Code Quality:** Clarity, efficiency, documentation, and adherence to best practices.
- **Report Quality:** Clarity, organization, and depth of analysis.
- **Teamwork and Collaboration:** Effective teamwork and communication within your group.
- **Code Explanation**: Ability to explain the code and the rationale behind design choices.

Phase II List of Deliverables:

1. Pdf file for the Report
2. .py file for the code (Models' implementation and EDA techniques)
3. A clear task division for each member of the team

Kaggle Competition will open in 15 April 2025 and will close in 29 April 2025

Phase II deadline (*submit all your project files*): 30 April 2025, 11:59 PM

Late Submission Policy:

Late submissions will be penalized according to the following schedule:

- **Up to 6 hours late:** 25% deduction in grade.
- **Up to 1 day late:** 50% deduction in grade.
- **More than 1 day late:** No grade will be assigned (0).

To ensure fairness and timely evaluation, please adhere to the submission deadlines.

Cheating policy

All work submitted in this project must be your own original work. Any form of academic dishonesty, including plagiarism, cheating, or unauthorized collaboration, is strictly prohibited. This includes the use of AI tools to generate code.
Such actions will result in severe consequences, including but not limited to having a ZERO grade on the project.